
Alpha Manufacturing Group (AMG)

Executive Memorandum

To: Board of Directors, Alpha Manufacturing Group (AMG)

From: Consultant, Leonard Yaw Mensah

Subject: ERP Transformation Recommendation

AMG should approach its ERP replacement as a controlled business change rather than just an IT project. To protect our 5 factory locations across the US and Canada, I recommend a phased cloud ERP rollout over a strict 12-month timeline. This strategy directly fixes our ongoing margin losses and engineering data errors while keeping factory floors running smoothly, and protect our highly profitable aftermarket parts and service business.

1. Replacement Strategy & 12-Month Plan

We must avoid an immediate "Big Bang" replacement where everything switches over on a single day. Because AMG builds highly customized machines, any single glitch in buying parts or sending designs to the factory floor will halt production and destroy our profits.

To safely hit the Board's 12-month deadline, the project will move through 4 distinct phases:

- **Phase 1: Clean & Align (Months 1–3):** Match our daily habits to the standard setup of the new software. Clean out the 20-year-old database and extract only active, current records.
- **Phase 2: Build & Connect (Months 4–8):** Set up the database for our US and Canadian branches. Build and test the automated link between the engineering software and the new ERP, and configure the core tracking modules for aftermarket parts and service history.
- **Phase 3: Test & Train (Months 9–11):** Have our top workers test the new system using old project files to make sure the math is correct. They then train the rest of the staff.
- **Phase 4: Launch & Support (Month 12):** Freeze the old system over a weekend to move final numbers. Go-live and keep technical experts on-site at all 5 factories for 30 days to fix user mistakes and other arising issues instantly.

Top Program Risks & Practical Mitigations:

- **Engineering Sync Failures:** Custom part lists transferring incorrectly from designers to the factory floor and other relevant departments, causing building delays. *Mitigation:* Set up an automatic validation gate that blocks broken designs before they reach the shop floor.
- **Employees Sticking to Old Spreadsheets:** Staff rejecting the new ERP and secretly continuing to run the company out of personal spreadsheets. *Mitigation:* Provide thorough team training, build simple dashboards that look like their old tools, and mandate that untracked jobs will not be paid.
- **Cross-Border Accounting Friction:** Math errors or delays when calculating US/Canadian dollar conversions and local taxes. *Mitigation:* Bring in external cross-border tax experts during Phase 1 to double-check the system's tax rules.
- **Scope Creep and Rising Costs:** Department heads demanding unique software customizations that delay the 12-month launch and blow past the budget. *Mitigation:* Enforce a strict "Fit-to-Standard" rule, we change our operational habits to match the software. A governance committee must reject all custom requests unless they are legally required.

2. Project Scope:

To guarantee a stable launch and meet the deadline, we must draw a hard line between what we need on day 1 versus what can wait.

- **Strictly In-Scope for Day 1:**
 - *Job Costing Sub-Ledger:* Unique Project IDs to track live material and labor spending, stopping our 80% margin losses instantly.
 - *Live Engineering Sync:* Auto-sending part lists directly from CAD tools to the shop floor to eliminate nightly sync delays.
 - *Aftermarket Parts & Service:* Full tracking of spare parts inventory and service history to protect core revenue.
 - *Basic Buying & Inventory:* Tracking raw materials (like steel) and core component inventory levels so builders have parts.
 - *Cross-Border Accounting:* US/Canada bookkeeping with automated dollar conversions and local taxes across our 5 sites.
 - *Manual Tablets & QA:* Touchscreens for workers to log hours and fill out basic, manual machine safety checklists.
- **Deferred (Out-of-Scope):**
 - *Factory Automation:* Automated machine tracking (IoT) and automated quality/defect analytics software.
 - *Advanced Logistics:* 3D warehouse shelf-mapping, RFID tracking, and customer-facing order progress portals.

- **Non-Critical Systems:** Deep CRM sales interaction histories and internal HR benefits or payroll processing software.

Data Migration Trade-Off:

We will explicitly exclude individual past invoices, timesheets, and closed project lines from the migration. Day 1 will only load active master files (current customers, active suppliers, current part catalogs) and open balances (unpaid invoices, open orders, and physical warehouse stock).

3. Keeping Data and Reports Running

- **The Data Lake Solution:** We will extract the entire 20-year history from the old system and save it in a secure, cheap cloud data storage lake, for example, S3 Glacier. The old ERP can then be safely turned off.
- **Uninterrupted Reporting:** We will link a simple reporting tool (like Power BI) to **both** the historical cloud data lake and the new live ERP database. When you run a 5-year trend report, the tool pulls data from both places seamlessly, keeping the new system clean and lightweight.

Month-End Continuity Plan:

The new system will launch precisely on Day 1 of a new fiscal month, never during a month-end closing window. The accounting team will complete the previous month's financial close entirely within the old system. Those final, certified numbers will be typed into the new cloud ERP as our opening baseline. For the first 30 days, the finance team will run **weekly practice closes** to catch and fix any cross-border currency errors well before the first official monthly deadline.

Conclusion

By using this simple, project-tracked, and engineering-linked plan, AMG can stop its margin losses, secure its aftermarket revenue, and stay within the 12-month limit without pausing factory operations. I look forward to working with AMG's leadership to start the project.